Ericsson PE Node BAS-PE-10 Basavanagudi POP Hardware/Environmental Fault Resolution SOP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_6259

Description

The PE node (Packet Engine) BAS-PE-10 at the Basavanagudi POP is experiencing a hardware/environmental fault as indicated by alarm code ALARM_6259. This fault is a result of ASIC memory parity errors or line card crashes, leading to a loss of power supply redundancy. The underlying cause of this issue lies in the hardware design of the PE node, specifically the memory management unit (MMU) of the ASIC, which is responsible for ensuring data integrity and consistency across the packet processing pipeline. In Ericsson's PE node architecture, the MMU employs a combination of error-correcting codes (ECC) and parity checks to detect and correct memory errors. However, in this instance, the redundancy mechanism is compromised due to a hardware failure, causing the system to enter a degraded state. Furthermore, the PE node's power supply redundancy loss exacerbates the situation, rendering the system unable to operate at optimal capacity. To resolve this issue, it is essential to identify and replace the faulty component(s), restore power supply redundancy, and verify the system's integrity.

Pre-requisite Checks

- Check: Verify the site location (Basavanagudi POP) and node ID (BAS-PE-10) to confirm the affected hardware and network topology., Task: Dispatch remote hands to the Basavanagudi POP for assistance.

- Check: Verify the redundant link status to ensure it is functioning correctly., Task: Check the link status using the command 'show redundancy' to confirm the redundant link is operational.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support | include ASIC, Expected Output: ASIC memory parity errors detected., Description: This command isolates the ASIC memory parity errors in the control plane by analyzing the system logs and identifying any errors related to the memory management unit (MMU). The output will indicate the presence of parity errors, which is a strong indication of the root cause.
```

```
Command: show redundancy | include Redundant, Expected Output:
Redundant link is operational., Description: This command isolates the
redundant link status in the data plane by checking the redundancy
mechanism. The output will confirm whether the redundant link is
operational or not, which is essential in identifying the root cause
of the issue.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace ASIC, Description: This step involves replacing the faulty ASIC with a new one. The command 'replace ASIC' will initiate the replacement process, and the system will automatically detect and configure the new ASIC.

Step 2: Step: Restore power supply redundancy, Command: redundancy configure, Description: This step involves restoring the power supply redundancy mechanism. The command 'redundancy configure' will reconfigure the system to use the redundant power supply, ensuring optimal system performance.

Post-Check Verification

```
Command: show redundancy | include Redundant, Expected Output:
Redundant link is operational., Description: This command verifies the
redundant link status in the data plane by checking the redundancy
mechanism. The output will confirm whether the redundant link is
operational or not, ensuring the system is functioning correctly.

Command: show tech-support | include ASIC, Expected Output: No ASIC
memory parity errors detected., Description: This command verifies the
ASIC memory parity errors in the control plane by analyzing the system
logs and identifying any errors related to the memory management unit
(MMU). The output will indicate the absence of parity errors,
confirming the issue is resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: configure replace, Description: This step involves restoring the original configuration before the replacement process. The command 'configure replace' will restore the original configuration, ensuring the system is in the same state as before the replacement process.

- Step: Replace the replaced component(s), Command: replace , Description: This step involves replacing the replaced component(s) with the original component(s). The command ' replace' will initiate the replacement process, and the system will automatically detect and configure the original component(s).